

CHRISTOPHER CAPIZZUTO

Software Engineer - Mobile, Full-Stack & Applied AI - builds web and mobile products, computer-vision features, and autonomous gameplay agents. Creator of **Dragapultist (50+ users)** and founder of **Brightboard Chess**.

Yonkers, NY | (347) 462-6803 | Chriszcodes@gmail.com | [GitHub.com/capisz](https://github.com/capisz) | [LinkedIn.com/in/chriszcodes](https://www.linkedin.com/in/chriszcodes) | capizzuto-portfolio.vercel.app | U.S. Citizen | English & Spanish

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Swift, SQL, HTML/CSS
 - **Web & mobile:** React, Next.js, React Native, Expo, SwiftUI, Electron, Tailwind CSS
 - **Backend & AI:** Node.js, Express, FastAPI, PyTorch, REST APIs, Claude API
 - **Data:** PostgreSQL/PostGIS, MongoDB, Firebase
 - **Delivery & testing:** Git, Docker, Kubernetes (local kind), GitHub Actions, Vercel, Playwright, Vitest, XCUITest, Firebase Emulator Suite
-

EXPERIENCE

Brightboard Chess - Founder · Current

- Founded an AI-focused chess education nonprofit to expand access for Black, Brown, and underserved students in New York City and beyond.
- Developing an instructional model that combines traditional chess teaching with AI-assisted learning and practice.
- **Finalist for the Andy Lerner Award for Chess Education Excellence.**

Independent Mobile & Full-Stack Developer · 2021 - Present

- Build and maintain web and mobile products, owning frontend development, APIs, databases, deployment, and production debugging.
- Implement authentication, third-party API integrations, and resilient payment workflows, including OAuth token handling and idempotent Stripe webhooks.

Northeast STEM Starter Academy (NSSA) - Head of Chess · Mount Vernon, NY | April 2026 - Present

- Lead three weekly chess classes for elementary and middle school students at a STEM education nonprofit; designed and taught a three-week camp across three skill levels.
- Created the operating plan and \$37,968 budget for a proposed Saturday Chess Academy expansion, including skill placement, staffing, and student readiness criteria.

SparrowUp - Software Engineer · May 2025 - April 2026

- Engineered motion-detection and tracking features for a soccer training app, including arm and leg segmentation to isolate limb movement for biomechanical analysis.
- Developed the analysis logic translating motion-capture output into actionable coaching feedback, quantifying seven biomechanical variables across four movement stages against professional benchmarks.
- Defined validation scenarios for tracking edge cases and measurement inconsistencies, helping the team identify motion-analysis defects before release.

North American Dental Group - Accountant · Ridge Hill office, Yonkers, NY | 2019 - 2023

- Maintained day-to-day bookkeeping records for a dental office, recording payments, expenses, and vendor invoices and organizing supporting financial documentation.
- Supported accounts payable and receivable, reconciled transactions against bank records, and helped resolve discrepancies and prepare routine financial reports.

SELECTED PROJECTS

Pokémon TCG Stockfish - Autonomous Gameplay & Kaggle Research · Python, PyTorch, TypeScript, FastAPI, React | In development · [GitHub](#)

- Developing a Python-based "**Stockfish for Pokémon TCG**": an analysis engine intended to evaluate positions, compare legal moves, and help players understand strategic decisions in a game with hidden information.
- Achieved a **top-16 finish** in [The Pokémon Company - PTCG AI Battle Challenge Strategy](#) on Kaggle; currently participating in the AI Battle Challenge Playground competition.
- Built a Python gameplay agent that ranks legal actions, plus a research pipeline connecting autonomous self-play, PyTorch training, saved checkpoints, and reproducible candidate evaluation.
- Built replay analysis with a FastAPI backend and React interface; separate experimental models from validated results while working toward reliable move grading and stronger play.

Dragapultist - Pokémon TCG Match Analyzer · Next.js, React, TypeScript, MongoDB, Electron | [Live](#) · [GitHub](#)

- Built and maintain a **downloadable beta app used by 50+ users**, turning Pokémon TCG Live exports into game history, win rates, and matchup insights.
- Designed indexed MongoDB schemas and aggregation pipelines for match statistics, with an Electron desktop build supporting local persistence and automated data ingestion.

PrizeCheck - Pokémon TCG Practice Trainer · Next.js, TypeScript, Firebase, Docker, Kubernetes, GitHub Actions | [Live](#) · [GitHub](#)

- Built a deck-import practice tool that simulates opening hands and prize cards, then scores recall by accuracy and speed.
- Implemented Google sign-in, saved decks, ownership-enforced Firestore rules, and offline-tolerant session sync with stable IDs to prevent duplicate records.
- Containerized the app and exercised pod replacement and rollback on a local kind cluster; configured CI with unit, security-rules, and browser tests.

Brightboard - Guided AI Chess Education · Next.js, React, TypeScript, PostgreSQL, Claude API | [GitHub](#)

- Built a fully guided AI chess education app with 30 beginner lessons, opening recall practice, puzzles, computer opponents, and game review; added a coach dashboard for assignments and student progress.
- Integrated kid-friendly Claude coaching grounded in engine-verified positions and puzzle solutions, with progressive hints and template fallbacks when AI is unavailable.

FunkFit - Adaptive Nutrition App · React Native, Expo, TypeScript, HealthKit, Fly.io | [GitHub](#)

- Built iOS and web clients from a shared TypeScript codebase with offline persistence and protected third-party credentials behind a backend proxy.
- Developed nutrition targets that use measured wearable energy when available, track the source of each estimate, and adjust goals to weight trends.

Pidge - NYC Parking Intelligence · SwiftUI, MapKit, Node.js, TypeScript, PostgreSQL/PostGIS | [GitHub](#)

- Built native and web map clients backed by spatial queries and normalized NYC curb data, including signs, meters, and street geometry.
- Designed conservative recommendations that exclude unknown and no-parking curbs; added backend, browser, and iOS test coverage.

Additional work: settle.ai, a room planner with a Three.js editor and browser-based floor-plan extraction; CareCation, a travel platform integrating Amadeus search and Stripe payments.

EDUCATION

Georgia Institute of Technology | M.S. Computer Science, AI specialization (incoming)

Codesmith | Full-Stack Software Engineering

Baruch College (CUNY) | B.B.A., Digital Marketing